Experimental Report 02

Time-Window Features and Feature Optimization

Remaining Useful Life prediction from multivariate sensor time series

Approach	Best MAE	Best R2	Feature set
time-window features	**13.23 cycles**	**0.817**	**336 features**

Time-window experiment setup

• The second experimental stage moved from single-cycle readings to fixed-size time windows, allowing the model to use recent sensor history rather than one instantaneous point.

• This representation increased computational cost: local training took approximately 4 minutes, while the earlier Random Forest version trained in about 7 seconds.

• The experiments compared window size, ignored sensor sets, key sensor behavior, and regression quality using MAE, RMSE and R2.

Window-size and sensor tests

Test	Window	Key sensor	Ignored sensors	MAE	RMSE	R2	Train, s	
1	50	11	None	15.016	20.500	0.757	166.4	
2	100	4	11	13.233	17.771	0.817	399.9	
3	30	4	11	19.652	26.214	0.602	132.8	
4	30	11	19, 3, 9, 5, 7, 10, 1	19.486	25.700	0.618	121.2	

Best tested configuration: window size 100 with sensor_11 excluded. The key sensor became sensor_4, and the model reached MAE 13.23, RMSE 17.77, and R2 0.817.

Observed sensor behavior

• With a 50-cycle window, sensor_11 remained the key indicator and the model improved relative to the earlier non-window baseline.

• When sensor_11 was excluded, sensor_4 became the key sensor, although an earlier hypothesis expected sensor_9 after its importance rate of 0.128949.

• This suggests that feature importance can change substantially after window construction and sensor exclusion.

Feature Optimization and Error Structure

Feature optimization result

An iterative feature-selection procedure was used to search for informative statistical blocks while controlling the search width. Candidate blocks were evaluated by their impact on MAE, allowing the system to remove features that did not reduce error or slowed the model.

MAE	RMSE	R2	Feature count
13.443413	**20.502046**	**0.756592**	**336**

Selected feature blocks

Optional blocks: rms, slope_change, last, first, median, iqr

Optional pairs: energy_acceleration, last_first, median_iqr

Top feature blocks

Rank	Feature block	Score
1	SLOPE	0.632081
2	MEAN_LAST_10	0.073060
3	MAX	0.050012
4	RANGE	0.011552

Prediction error structure

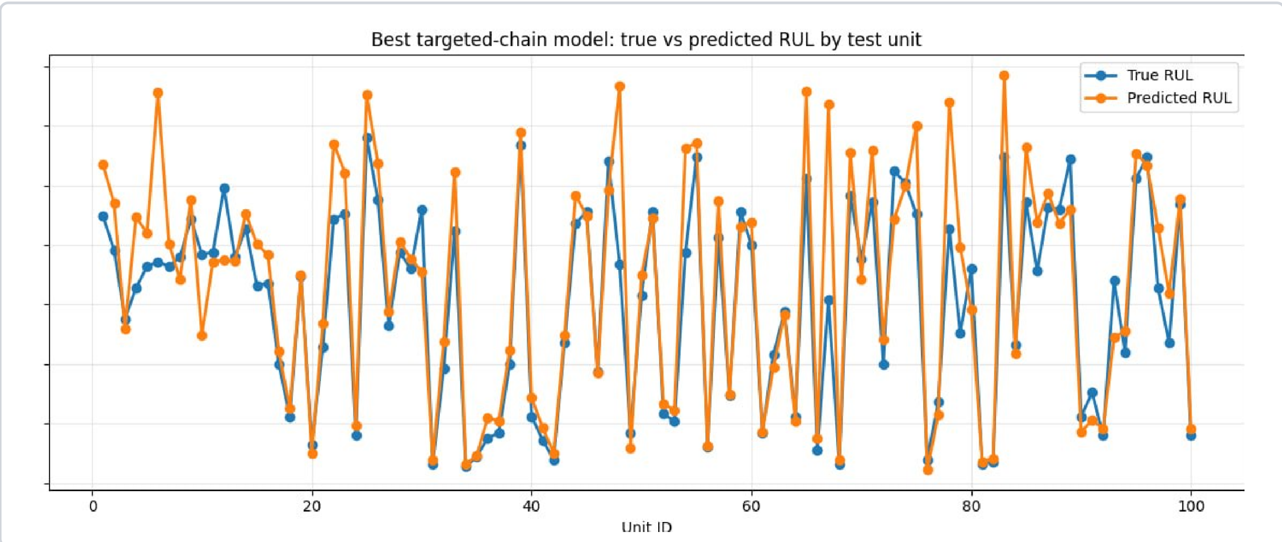

Figure 1. True vs predicted RUL by test unit. The comparison indicates that the largest visible deviations are concentrated mainly around initial/high-RUL values, while the model often follows the relative structure of the target signal.

Report summary

Representation	fixed-size time windows over recent sensor history
Best test result	window 100, sensor_11 excluded: MAE 13.23, R2 0.817
Feature optimization	336-feature set with slope-related blocks as strongest signals
Error pattern	largest visible deviations mainly at initial/high-RUL values